Assignment 1

Name: Ruhan Garg
Batch: 3Q31
Roll No: 1204170291

1. Leap Year

```
import java.util.Scanner;

class LeapYear {
 public static void main(String[] args) {
 Scanner sc = new Scanner(System.in);
 System.out.print("enter year: ");
 int year = sc.nextInt();

 if (year % 4 == 0 && (year % 100 != 0 || year % 400 == 0))
System.out.println("leap year");
 else
 System.out.println("not a leap year");  }
}
```

2. Check Alphabet

```
import java.util.Scanner;

class CheckAlphabet {
 public static void main(String[] args) {
 Scanner sc = new Scanner(System.in);
 System.out.print("enter a character: ");
 char ch = sc.next().charAt(0);

 if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
System.out.println("alphabet");
 else
 System.out.println("not alphabet");
 }
}
```

3. Factorial

```
import java.util.Scanner;

class Factorial {
 public static void main(String[] args) {
 Scanner sc = new Scanner(System.in);
 System.out.print("enter n: ");
```

```
 int n = sc.nextInt();

 int fact = 1;
 for (int i = 1; i <= n; i++) {  fact
= fact * i;
 }
 System.out.println("factorial = " + fact);  }
}
```

4. Fibonacci Series

```
import java.util.Scanner;

class Fibonacci {
 public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("how many terms: ");  int n
= sc.nextInt();

 int a = 0, b = 1;
 for (int i = 0; i < n; i++) {
 System.out.print(a + " ");  int c = a
+ b;
 a = b;
 b = c;
 }
 }
}
```

5. GCD

```
import java.util.Scanner;

class GCD {
 public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("enter two numbers: ");  int a
= sc.nextInt();
 int b = sc.nextInt();

 while (b != 0) {
 int r = a % b;
 a = b;
 b = r;
 }
 System.out.println("gcd = " + a);  }
}
```

6. LCM

```
import java.util.Scanner;
```

```
class LCM {
 public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("enter two numbers: ");  int a
= sc.nextInt();
 int b = sc.nextInt();

 int m = a, n = b;
 while (n != 0) {
 int r = m % n;
 m = n;
 n = r;
 }
 int lcm = (a * b) / m;
 System.out.println("lcm = " + lcm);  }
}
```

7. Count Digits

```
import java.util.Scanner;

class CountDigits {
 public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("enter a number: ");  int n
= sc.nextInt();

 int count = 0;
 while (n != 0) {
 n = n / 10;
 count++;
 }
 System.out.println("digits = " + count);  }
}
```

8. Reverse a Number

```
import java.util.Scanner;

class ReverseNumber {
 public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("enter a number: ");  int n
= sc.nextInt();

 int rev = 0;
 while (n != 0) {
 int d = n % 10;
 rev = rev * 10 + d;
 n = n / 10;
```

```
 }
 System.out.println("reversed = " + rev);  }
}
```

9. Power of a Number

```
import java.util.Scanner;

class Power {
 public static void main(String[] args) {  Scanner sc
= new Scanner(System.in);  System.out.print("enter
base and exponent: ");  int base = sc.nextInt();
 int exp = sc.nextInt();

 int result = 1;
 for (int i = 1; i <= exp; i++) {
 result = result * base;
 }
 System.out.println("result = " + result);  }
}
```

10. Palindrome

```
import java.util.Scanner;

class Palindrome {
 public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("enter a number: ");  int n
= sc.nextInt();

 int temp = n;
 int rev = 0;
 while (temp != 0) {
 int d = temp % 10;
 rev = rev * 10 + d;
 temp = temp / 10;
 }
 if (n == rev)
 System.out.println("palindrome");  else
 System.out.println("not palindrome");  }
}
```